Rohit Kumar

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Summary

Data Scientist & AI-ML Engineer specializing in Generative AI architectures and production-grade ML pipelines. Expert in developing RAG systems using LangChain, predictive analytics with XGBoost, and distributed data processing via PySpark. Proven success in optimizing insurance data ecosystems to achieve 35%+ improvements in retrieval and forecasting accuracy.

Technical Skills

Languages: Python (Pandas, NumPy, Scikit-learn), SQL, Java, C++, R

AI/ML & GenAI: Large Language Models (LLMs), RAG, LangChain, ChromaDB, XGBoost, Prophet

Data Engineering: PySpark, ETL Pipelines, PostgreSQL, Docker, Apache Airflow, Jenkins

Experience

Go-Digit Life Insurance Ltd. June 2024 – Present
Data Scientist (ML Engineering) *Pune, India*

- Spearheaded **SmartDoc AI**, a RAG-based Generative AI system using **LangChain** and **OpenAI** to automate knowledge extraction from 1,000+ insurance policy documents.
- Engineered a production-ready **Churn Prediction Pipeline**, utilizing **XGBoost** and **SMOTE** to identify high-risk customer segments with 92% accuracy.
- Deployed a **Demand Forecasting engine** using **Facebook Prophet**, reducing operational forecasting errors by 18% through automated seasonal trend analysis.
- Optimized feature engineering workflows by migrating legacy insurance schemas to **PostgreSQL**, improving data retrieval latency by 35% and streamlining MLOps with **Docker**.

Projects

SmartDoc AI: RAG-based Generative AI *LangChain, OpenAI, ChromaDB, Python*

- Architected a Retrieval-Augmented Generation (RAG) framework using LangChain to enable semantic search and real-time QA on technical documentation.
- Implemented optimized text chunking strategies and vector embeddings to reduce model hallucinations by 40%.
- Evaluated model performance using **RAGAS** (RAG Assessment) metrics, achieving 85%+ faithfulness and relevancy across domain-specific insurance queries.

Bank Customer Churn Prediction & Analytics *XGBoost, SMOTE, Python, SQL*

- Developed a binary classification model to predict attrition risk for 10k+ customers, achieving 92% accuracy through SMOTE class balancing.
- Derived data-driven retention strategies that improved marketing efficiency by a projected 20%.

Demand Forecasting & Supply Chain Optimization *Facebook Prophet, Python, Pandas*

- Built a time-series forecasting engine utilizing Facebook Prophet to predict seasonal trends and inventory demand.
- Reduced forecasting error (MAPE) by 18% through automated parameter optimization.

Education

C. V. Raman Global University 2020 – 2024
B.Tech in Computer Science and Engineering (CGPA: 8.7) *Bhubaneswar*

Certifications & Achievements

IEEE Publication: "Load Balancing in Cloud Computing" (2024) — View on IEEE Xplore

PCAP Certified Python Developer: Advanced proficiency in modular data-centric programming.

Competitive Programming: 5-star coder on HackerRank; 200+ problems solved on LeetCode/GFG.